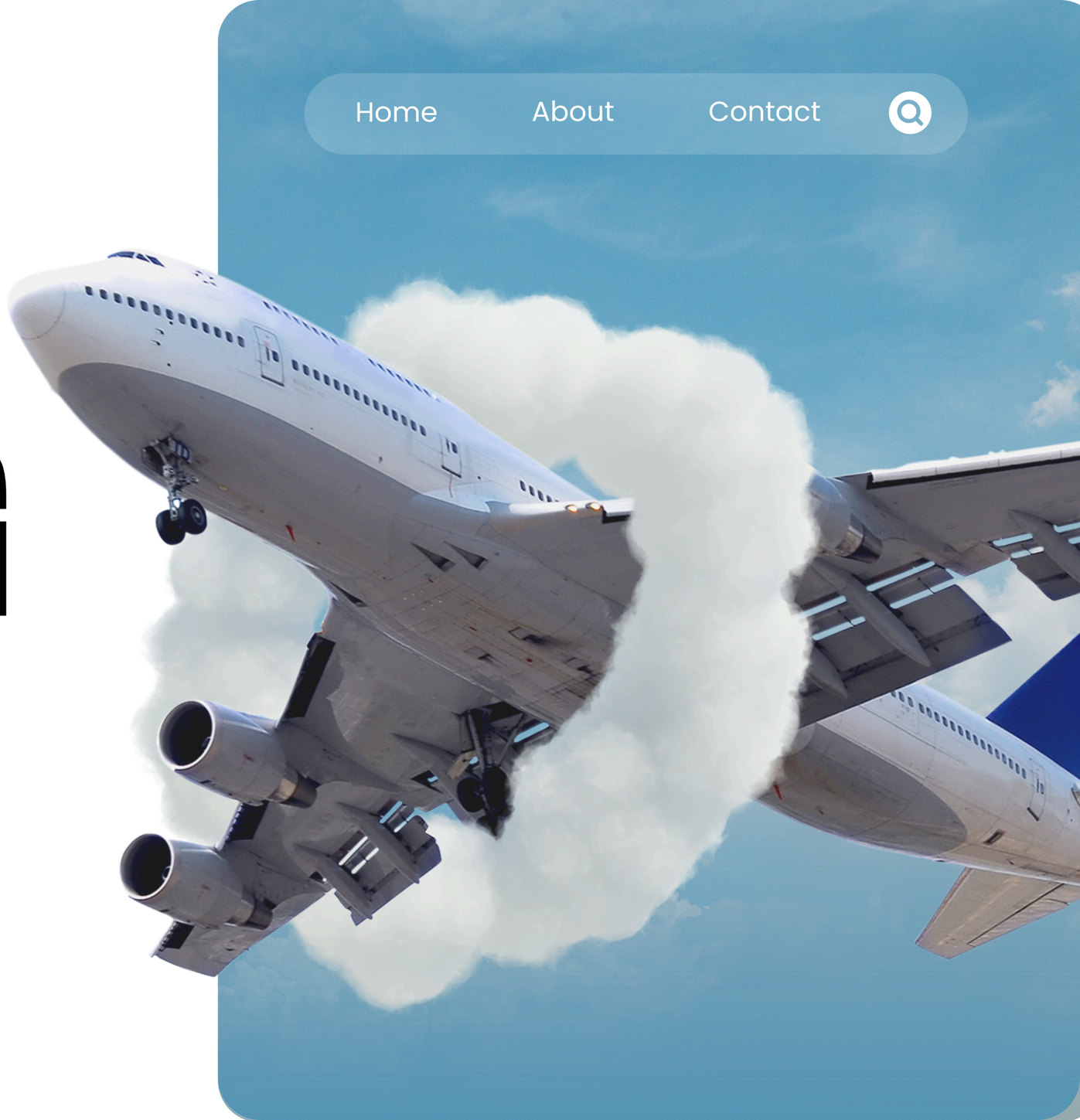
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AIRLINE BOOKING



Every great adventure begins with a single step
and we are here to guide that first step.

Get Started

INTRODUCTION

Objective

To analyze airline booking patterns, customer preferences, and operational performance using Excel, Python, and SQL.

Dataset

25,000 booking records with passenger, trip, and booking details.

Tools Used

Excel, Python (Pandas, Numpy, Matplotlib, Seaborn), SQL Server.

DATASET OVERVIEW

- Part 1: Data (Sample)

	A	B	C	D	E	F	G	H	I
1	num_passengers	sales_channel	trip_type	purchase_lead	length_of_stay	flight_hour	flight_day	route	booking_origin
2	2	Internet	RoundTrip	262	19	7	Sat	AKLDEL	New Zealand
3	1	Internet	RoundTrip	112	20	3	Sat	AKLDEL	New Zealand
4	2	Internet	RoundTrip	243	22	17	Wed	AKLDEL	India
5	1	Internet	RoundTrip	96	31	4	Sat	AKLDEL	New Zealand
6	2	Internet	RoundTrip	68	22	15	Wed	AKLDEL	India
7	1	Internet	RoundTrip	3	48	20	Thu	AKLDEL	New Zealand
8	3	Internet	RoundTrip	201	33	6	Thu	AKLDEL	New Zealand
9	2	Internet	RoundTrip	238	19	14	Mon	AKLDEL	India
10	1	Internet	RoundTrip	80	22	4	Mon	AKLDEL	New Zealand
11	1	Mobile	RoundTrip	378	30	12	Sun	AKLDEL	India
12	2	Internet	RoundTrip	185	25	14	Tue	AKLDEL	United Kingdom
13	1	Internet	RoundTrip	8	43	2	Sat	AKLDEL	New Zealand
14	4	Internet	RoundTrip	265	24	19	Mon	AKLDEL	New Zealand
15	1	Internet	RoundTrip	185	17	14	Fri	AKLDEL	India
16	1	Internet	RoundTrip	245	34	4	Tue	AKLDEL	New Zealand
17	1	Internet	RoundTrip	192	18	14	Thu	AKLDEL	India
18	1	Internet	RoundTrip	259	37	6	Sun	AKLDEL	India
19	1	Internet	RoundTrip	19	29	10	Sat	AKLDEL	New Zealand
20	1	Internet	RoundTrip	67	155	8	Sun	AKLDEL	New Zealand
21	1	Internet	RoundTrip	351	17	3	Sun	AKLHGH	China
22	2	Internet	CircleTrip	228	29	23	Wed	AKLHND	New Zealand
23	1	Internet	RoundTrip	21	88	9	Wed	AKLICN	South Korea
24	2	Internet	RoundTrip	23	19	8	Mon	AKLICN	New Zealand

POTENTIAL QUESTIONS

1. Booking Trends
   - Which sales channel has the highest number of bookings?
2. Customer Behavior
   - What percentage of customers request extra baggage, preferred seats, or meals?
3. Timing Analysis
   - What is the average purchase lead time?
4. Geographic Analysis
   - Which countries have the most bookings?
5. Performance Metrics
   - Average passengers, flight durations.



DATA CLEANING

	A	B	C	D	E	F	G	H	I	J
1	num_passengers	sales_channel	trip_type	purchase_lead	purchase_lead_group	length_of_stay	flight_hour	day_period	flight_day	day_type
2	1	Online	RoundTrip	47	More than 31+ days	17	23	Evening	Wed	Weekday
3	1	Online	RoundTrip	140	More than 31+ days	17	6	Morning	Sat	Weekend
4	1	Online	RoundTrip	82	More than 31+ days	17	12	Afternoon	Tue	Weekday
5	1	Online	RoundTrip	27	8–30 days	17	3	Night	Fri	Weekday
6	1	Online	RoundTrip	63	More than 31+ days	17	12	Afternoon	Fri	Weekday
7	1	Online	RoundTrip	71	More than 31+ days	17	6	Morning	Wed	Weekday
8	1	Online	RoundTrip	27	8–30 days	17	1	Night	Fri	Weekday
9	1	Online	RoundTrip	48	More than 31+ days	17	2	Night	Fri	Weekday
10	1	Online	RoundTrip	27	8–30 days	17	3	Night	Fri	Weekday
11	1	Online	RoundTrip	0	0–7 days	17	1	Night	Sun	Weekend
12	1	Online	RoundTrip	77	More than 31+ days	17	7	Morning	Tue	Weekday
13	1	Online	RoundTrip	64	More than 31+ days	17	10	Morning	Sat	Weekend
14	1	Online	RoundTrip	20	8–30 days	17	12	Afternoon	Fri	Weekday
15	1	Online	RoundTrip	22	8–30 days	17	9	Morning	Wed	Weekday
16	1	Online	RoundTrip	8	8–30 days	17	8	Morning	Mon	Weekday
17	1	Online	RoundTrip	67	More than 31+ days	17	10	Morning	Sun	Weekend
18	1	Online	RoundTrip	64	More than 31+ days	17	10	Morning	Wed	Weekday
19	1	Online	RoundTrip	126	More than 31+ days	17	7	Morning	Wed	Weekday
20	1	Online	RoundTrip	53	More than 31+ days	17	7	Morning	Sun	Weekend
21	1	Online	RoundTrip	206	More than 31+ days	17	10	Morning	Mon	Weekday
22	1	Online	RoundTrip	42	More than 31+ days	17	6	Morning	Wed	Weekday
23	1	Online	RoundTrip	8	8–30 days	17	8	Morning	Sun	Weekend
24	1	Online	RoundTrip	96	More than 31+ days	17	1	Night	Fri	Weekday
25	1	Online	RoundTrip	79	More than 31+ days	17	6	Morning	Mon	Weekday
26	1	Online	RoundTrip	62	More than 31+ days	17	2	Night	Tue	Weekday
27	1	Online	RoundTrip	24	8–30 days	17	0	Night	Tue	Weekday
28	1	Online	RoundTrip	44	More than 31+ days	17	16	Afternoon	Wed	Weekday
	1	Online	RoundTrip	35	More than 31+ days	17	3	Night	Sat	Weeker

Checks Performed

- ✅ Duplicate rows – none found
- ✅ Data type validation – all correct
- ✅ Outliers – none detected

Actions Taken

- Missing values (no set → “Egypt”)
- Converted Boolean fields (0/1 → Yes/No)
- Standardized text categories (consistent spelling)

DATA TRANSFORMATION



New Columns Created:

- trip_duration_category → Short / Medium / Long
- day_type → Weekend / Weekday
- day_period → Morning / Afternoon / Evening / Night
- purchase_lead_group → 0–7 days / 8–30 days / More than 31+ days
- booking → Complete / Non-Complete



WORKFLOW

Pipeline:

- Excel – Initial inspection, pivot tables
- Python (Pandas) – Cleaning, transformations, export
- SQL Server – Analytical queries, aggregations
- Python (Matplotlib/Seaborn) – Visualizations

Excel → Python → SQL → Python Viz

EXCEL ANALYSIS

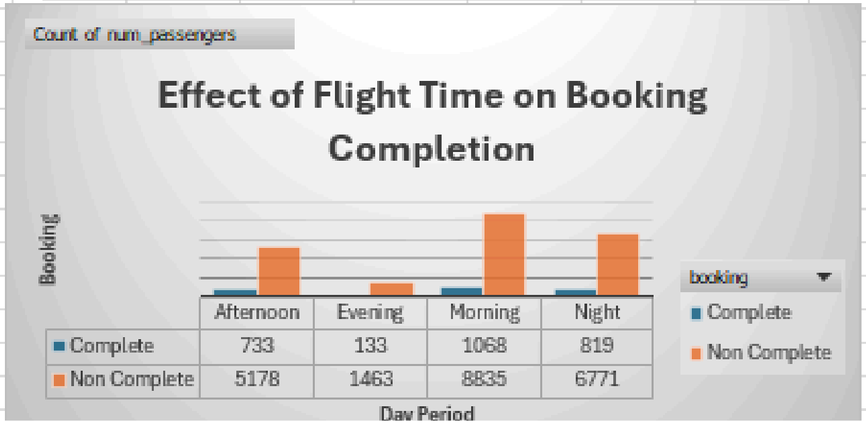
- Pivot Tables: Bookings by channel, trip type, country
- Charts: Colum, Pie, Line

Count of booking_complete	Column Labels			
Row Labels	CircleTrip	OneWay	RoundTrip	Grand Total
Offline	2	9	2060	2071
Online	69	262	22598	22929
Grand Total	**71**	**271**	**24658**	**25000**

Count of num_passengers	Column Labels		
Row Labels	Complete	Non Complete	Grand Total
Afternoon	733	5178	5911
Evening	133	1463	1596
Morning	1068	8835	9903
Night	819	6771	7590
Grand Total	**2753**	**22247**	**25000**

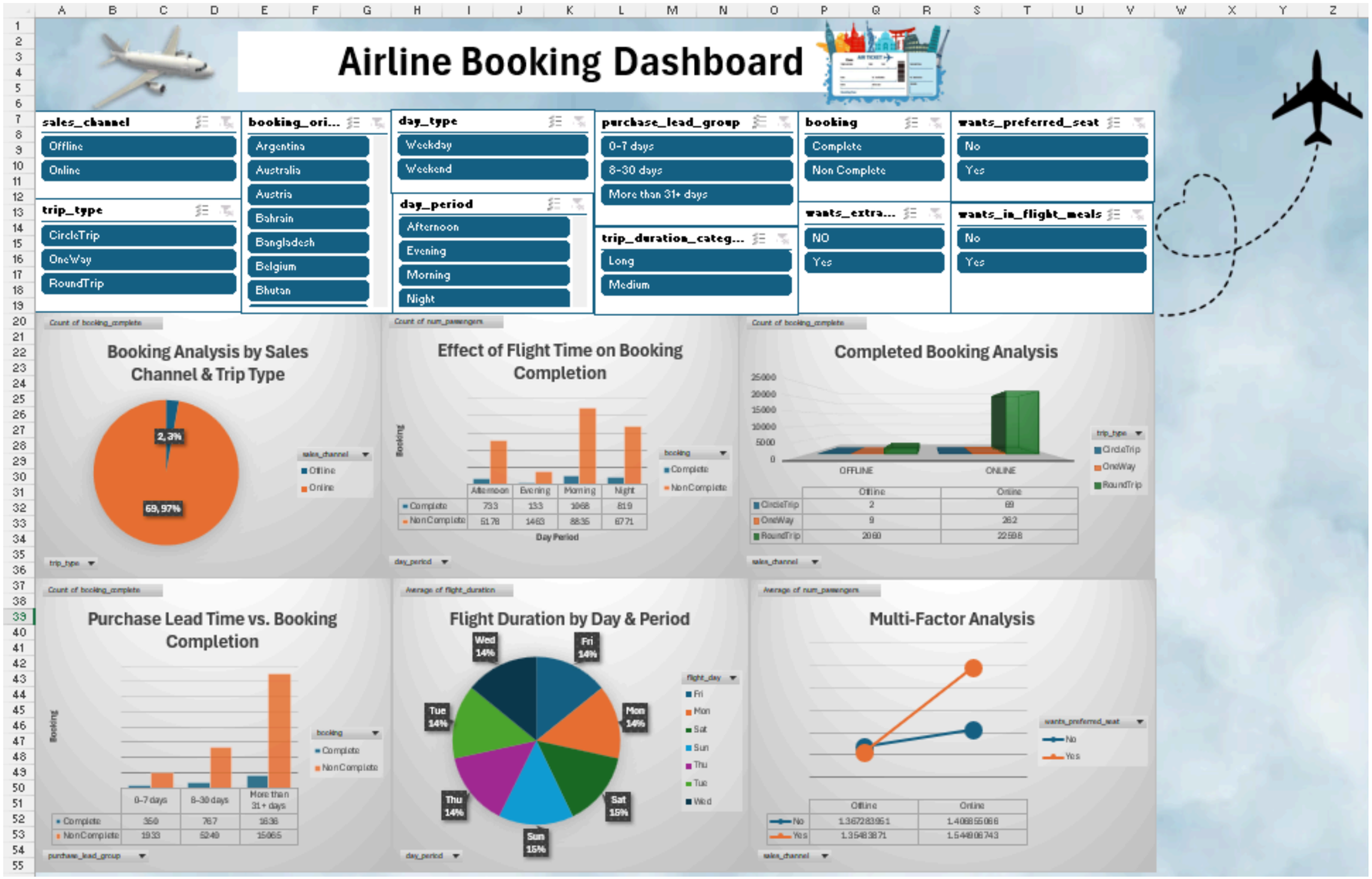
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DASHBOARD

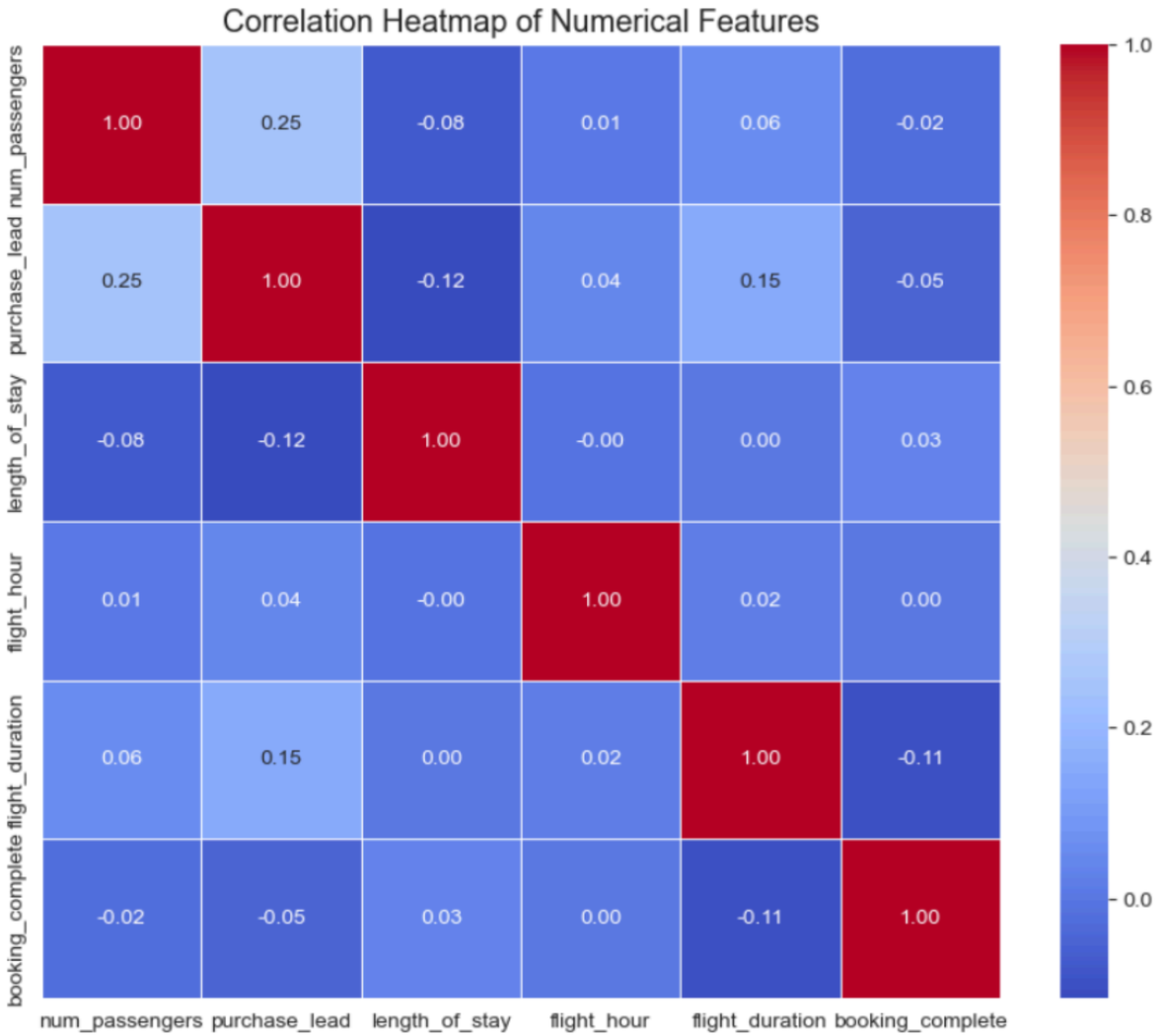


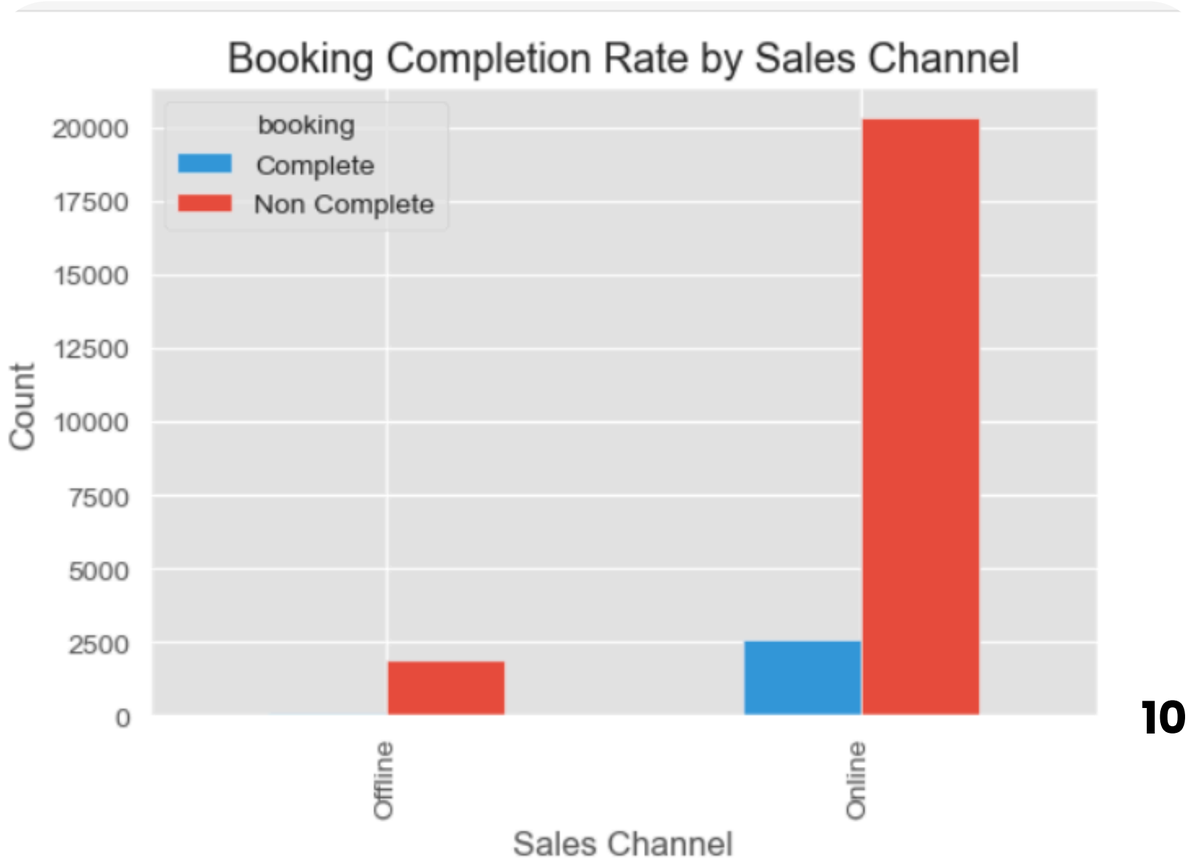
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PYTHON ANALYSIS

Examples:

- Bar chart: Sales channel distribution
- Pie chart: Trip type share
- Heatmap: Flight day vs. flight hour bookings
- Boxplot: Flight duration per trip type





SQL ANALYSIS

```sql
-- Booking Trends
-- 1.1
SELECT
    sales_channel,
    COUNT(*) AS booking_count
FROM [dbo].[Sheet1$]
GROUP BY [sales_channel]
ORDER BY booking_count DESC;
```

```sql
--4.2
SELECT
    booking_origin,
    AVG(flight_duration) AS avg_flight_duration
FROM [dbo].[Sheet1$]
GROUP BY booking_origin
ORDER BY avg_flight_duration DESC;
```

KEY INSIGHTS



- Channel: Majority of bookings are (Online → 91.7%).
- Customer Preferences: 78.4% request baggage, 48.5 % meals, 30.9% seats.
- Timing: Most flights depart on [Monday].
- Geography: Top origin countries: [Australia, Malaysia, South Korea].
- Performance: Avg. flight duration (Medium = 5.7 hrs, Long = 8.42 hrs ,Avg. passengers = 1.4 ~ 1–2 passengers).

CONCLUSION

- The project demonstrated a full data pipeline from raw dataset to business insights.
- Applied multi-tool approach for flexibility & cross-verification.

TEAM MEMBER

- Yara Adel Elshamly
- Mennatullah Mohamed Mahrous
- Kenzy Bahaa Ghoneim

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THANK YOU

Eng.Ali Abdullah